BMI585 — Kalman Filter Project Report

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Abstract

Cardiorespiratory coupling during sleep is a potential marker of autonomic regulation and, when disrupted, may reflect early cardiovascular dysregulation. In this project, I used overnight polysomnography (PSG) data from the HSP cohort to study whether respiration helps explain heart rate (HR) dynamics beyond what HR explains from its own history. I implemented and compared two 1D Kalman filter models on the same 30 minute segment: (A) a self-predictive AR(1) state space model for HR, and (B) an ARX(1) extension that adds a respiration driver derived from nasal airflow (PTAF) as an exogenous input (with an optional lag). Models were tuned via a small grid search and evaluated using one step ahead prediction error and filtered state error, along with innovation diagnostics (time series and histogram). On this segment, Model A achieved lower error ($\text{RMSE}_{\text{pred}} = 1.250$, $\text{RMSE}_{\text{filt}} = 0.771$) than Model B ($\text{RMSE}_{\text{pred}} = 1.578$, $\text{RMSE}_{\text{filt}} = 1.221$), and the tuned respiration coefficient in Model B selected a small negative coupling to $b \approx -0.2$ with best lag $= 10$. These results suggest that, for the chosen preprocessing and window, PTAF did not provide additional predictive structure for HR beyond a simple persistent AR(1) baseline model. I discuss likely reasons (driver construction, lag/feature mismatch, and model simplicity) and next steps toward richer coupling models.

1 Problem Definition

Cardiovascular diseases (CVDs) can develop silently, and subtle irregularities in autonomic regulation may appear before overt symptoms. During sleep, ECG-related measures (like HR) and respiration provide a window into cardiorespiratory dynamics. The goal of this project is to test whether adding respiration information improves a model of HR dynamics and whether innovation properties can serve as interpretable diagnostics.

A Kalman filter provides recursive estimation of a latent physiological state in the presence of process and measurement noise. Under correct modeling assumptions, the innovation sequence should be approximately white and Gaussian with predictable variance; structured or heteroscedastic innovations may indicate either model mismatch or physiologic irregularity.

Concretely, I compare:

- **Model A (self-predictive):** HR is modeled from its own history.
- **Model B (respiration-driven):** HR is modeled from its own history plus a respiration driver derived from PTAF.

2 Dataset Summary

This project uses the HSP overnight PSG cohort stored in HDF5 format. The cohort includes approximately $119{,}000$ overnight sessions sampled at 200 Hz with multiple physiological channels and expert annotations. Each session contains many signals (including ECG, HR, and PTAF) and annotation channels (sleep stage and arousal). In the full-night records, channels typically have $N \approx 6.09 \times 10^6$ samples at 200 Hz (about 8.46 hours). Also the Dataset is available at : this directory: "/opt/scratchspace/qli50/HSP"

For experiments and visualization, I used the example subject/window shown in my notebook output (cohort I0004; file name shown in the plot titles) and focused on:

- **Cardiac observable:** HR (from `signals/hr`) downsampled to 1 Hz.
- **Respiration proxy:** PTAF (nasal airflow; `signals/ptaf`) transformed into a 1 Hz respiration driver.

3 Preprocessing

Let k index seconds (1 Hz). From the raw 200 Hz signals, I constructed:

- A 1 Hz HR sequence y_k over a 30-minute window (length 1800 samples).
- A 1 Hz respiration driver u_k derived from PTAF and standardized to make the scale comparable to HR modeling.

Figure 1 summarizes the raw signals (HR and PTAF) and their processed 1 Hz versions used by the filters.

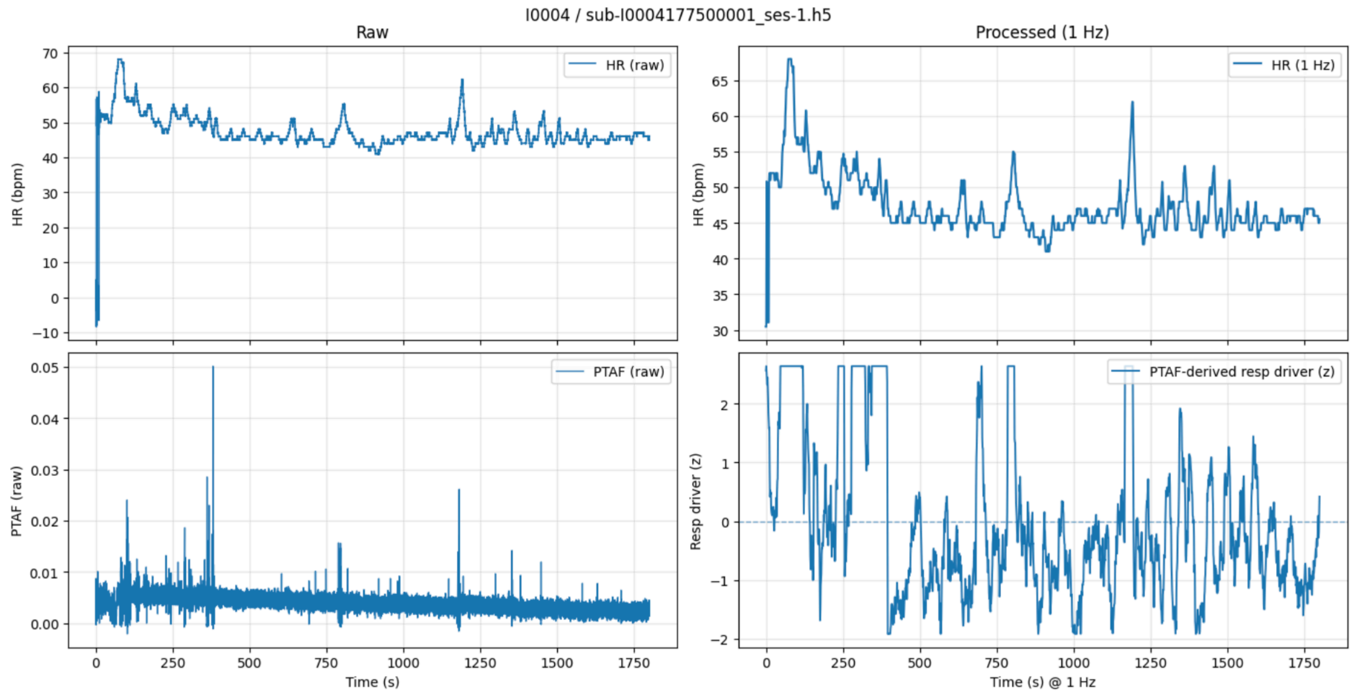

Figure 1: Raw HR and PTAF (left) and processed 1 Hz HR plus standardized respiration driver derived from PTAF (right). Replace filename with your saved figure.

4 State-Space Model Designs

4.1 Common observation model

Both models treat the observed 1 Hz HR as a noisy measurement of an underlying latent HR state:

$$y_k = x_k + v_k, \tag{1}$$
$$v_k \sim \mathcal{N}(0, R), \tag{2}$$

where:

- x_k is the latent (noise-reduced) HR state at second k,
- R is the measurement noise variance (sensor + preprocessing noise).

4.2 Model A: AR(1) Kalman filter (1D)

Model A assumes HR is slowly varying and approximately persistent:

$$x_{k+1} = a\, x_k + w_k, \tag{3}$$
$$w_k \sim \mathcal{N}(0, Q), \tag{4}$$

where:

- a is the AR(1) persistence coefficient (typically close to 1 for slow drift),
- Q is the process noise variance (how much HR is allowed to change from one second to the next beyond the AR prediction).

Connection to a continuous-time intuition (optional). If one starts from a continuous-time drift model

$$\frac{dx}{dt} = \alpha x(t) + w(t),$$

then a simple Euler discretization with step Δt gives

$$x_{k+1} \approx x_k + \alpha \Delta t \, x_k + w_k = (1 + \alpha \Delta t) x_k + w_k,$$

so the discrete-time coefficient corresponds to $a = 1 + \alpha \Delta t$. With 1 Hz sampling, $\Delta t = 1$, hence $a = 1 + \alpha$.

4.3 Model B: ARX(1) Kalman filter (1D + respiration input)

Model B adds a respiration driver u_k (from PTAF), allowing respiration to influence HR:

$$x_{k+1} = a \, x_k + b \, u_{k-\ell} + w_k, \tag{5}$$

$$w_k \sim \mathcal{N}(0, Q), \tag{6}$$

with the same observation model $y_k = x_k + v_k$. Here:

- b is the coupling strength from respiration driver to HR,
- ℓ is an integer lag (seconds) allowing delayed respiratory influence.

Interpretation goal. If respiration explains additional structure in HR dynamics, we expect Model B to reduce innovation magnitude/variance relative to Model A and to yield less structured innovations over time.

5 Kalman Filtering and Innovation

At each second k, the Kalman filter produces:

- a **one-step prediction** $\hat{x}_{k|k-1}$,
- a **filtered estimate** $\hat{x}_{k|k}$ after seeing y_k,
- an **innovation** (prediction residual)

$$e_k = y_k - \hat{x}_{k|k-1}.$$

Innovation plots and histograms provide diagnostics: large bursts, heavy tails, or nonstationary variance can indicate model mismatch (or true physiological events).

6 Parameter Tuning via Grid Search

Because Q and R strongly control smoothness vs. responsiveness, I tuned parameters by a small grid search:

- **Model A:** search over (a, Q, R) and choose the setting minimizing the RMSE of one-step predictions (and/or filtered estimate) against the observed HR.
- **Model B:** search over (ℓ, b, Q, R) (with a also tuned or held near the tuned value) and choose the setting minimizing the same metric.

This is a pragmatic approach to obtain stable behavior for a short segment; more principled alternatives include likelihood-based estimation or EM for state-space models.

7 Results

7.1 Quantitative comparison

Table 1 summarizes the final tuned parameters and errors computed on the same 30-minute HR–PTAF window.

Model	a	b	lag ℓ	Q	R	RMSE (pred / filt)
A: AR(1)	0.9995	–	–	0.1878	0.9392	1.250 / 0.771
B: ARX(1)	0.9995	-0.2	10	0.0939	1.8784	1.578 / 1.221

Table 1: Final comparison on the HR–PTAF segment. "pred" refers to one-step-ahead prediction error, and "filt" refers to filtered estimate error, both against observed HR.

7.2 Model A behavior

Figure 2 shows that Model A tracks the observed HR closely while maintaining smoothness. After an initial transient, the innovation time-series stays mostly within a stable band with occasional excursions; the innovation histogram is centered near zero with moderate tails, suggesting a reasonable (though not perfect) match for this segment.

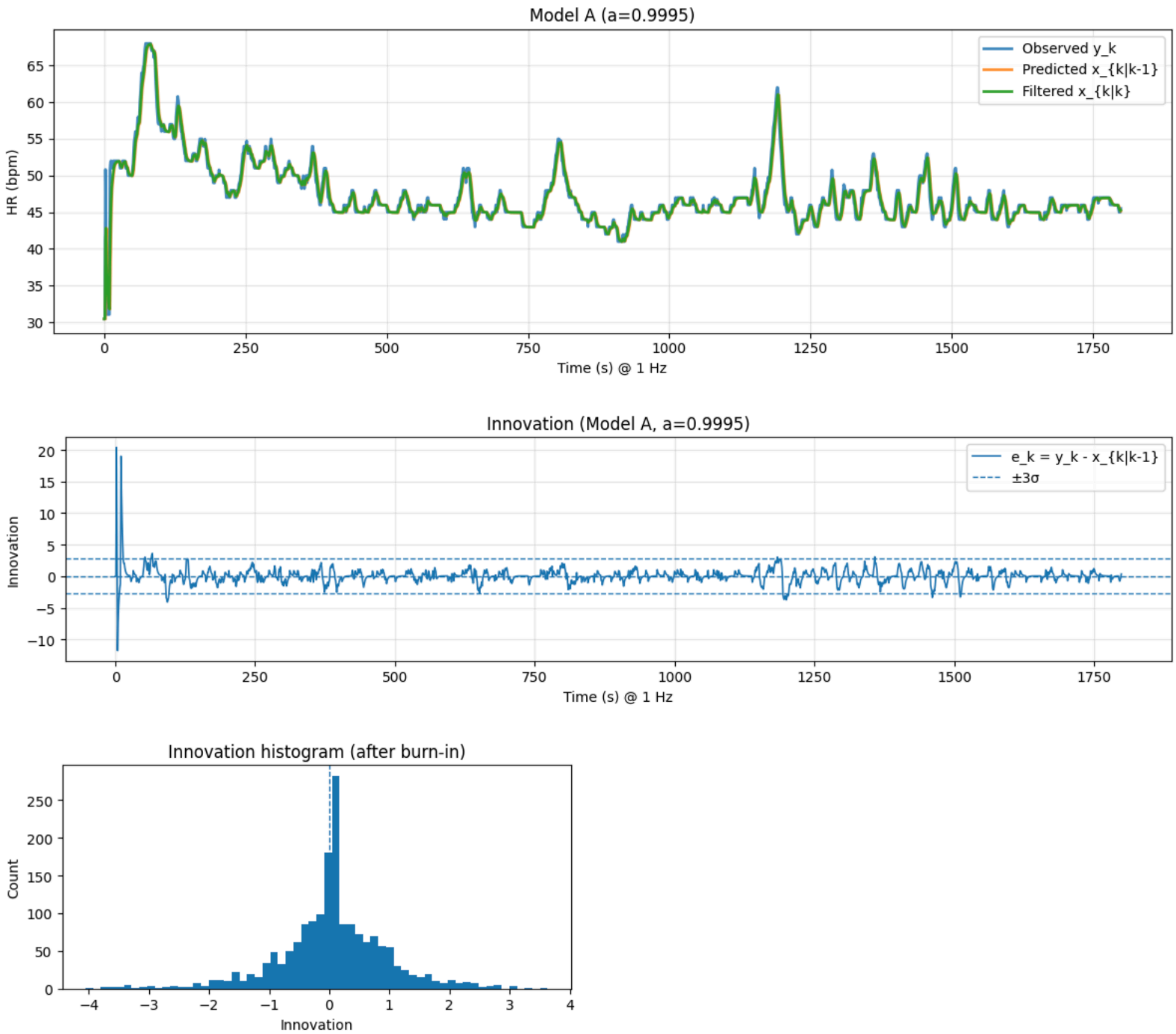

Figure 2: Model A: observed HR vs. predicted $\hat{x}_{k|k-1}$ and filtered $\hat{x}_{k|k}$, plus innovation diagnostics (time-series and histogram). Replace filename with your saved figure.

7.3 Model B behavior

Figure 3 shows similar tracking performance, but the tuned coupling selected a negative coefficient ($b = -0.2$) with best lag $\ell = 10$. Quantitatively, both RMSE metrics are worse than Model A. The innovation diagnostics do not show a clear reduction in variance or structure compared to Model A for this respiration driver.

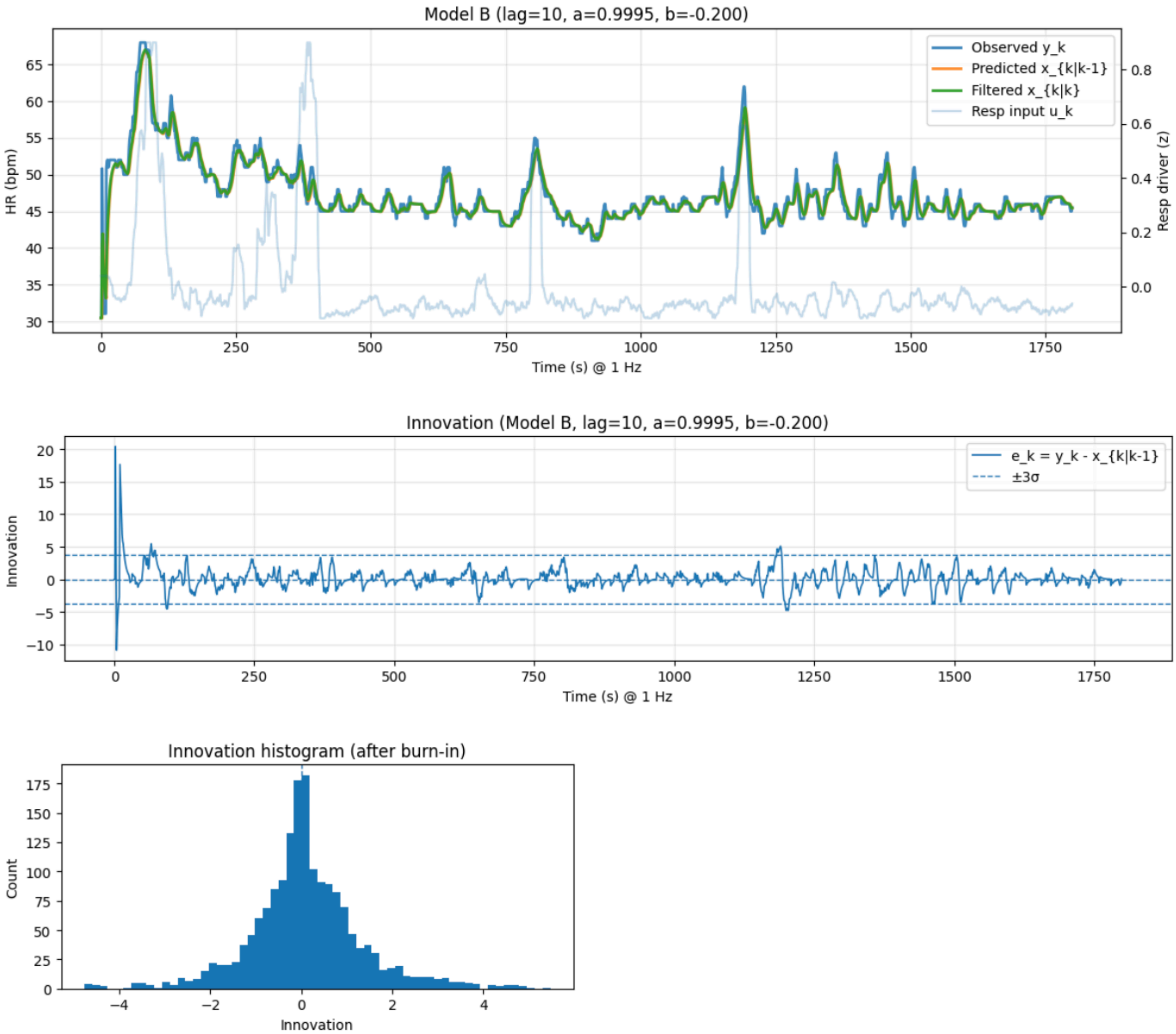

Figure 3: Model B: HR tracking and innovation diagnostics with respiration driver input. Replace filename with your saved figure.

8 Interpretation and Discussion

8.1 Which model performed better?

On this segment, **Model A performed better** by both error metrics:

$$\text{RMSE}_{\text{pred}} : 1.250 \text{ (A)} < 1.578 \text{ (B)}, \qquad \text{RMSE}_{\text{filt}} : 0.771 \text{ (A)} < 1.221 \text{ (B)}.$$

Interpretation of the tuned respiration term (b) and lag (ℓ). In Model B the state dynamics are

$$x_{k+1} = a\,x_k + b\,u_{k-\ell} + w_k,$$

where u_k is the PTAF-derived respiration driver (standardized) and ℓ is an integer lag in seconds.

The grid search selected $\ell = 10$ and $b = -0.2$. This means the filter is using the respiration driver from *10 seconds earlier* to adjust the HR state prediction at time $k+1$. Because $b < 0$, the effect is *inverse*: when the respiration driver is above its mean (e.g., $u_{k-10} > 0$ after z-scoring), the model predicts a slightly *lower* latent HR one step later, whereas when the driver is below its mean ($u_{k-10} < 0$), the model predicts a slightly *higher* latent HR.

The magnitude is linear and small: a one-standard-deviation increase in the driver (i.e., $u_{k-10} = +1$) changes the predicted state by approximately $b\, u_{k-10} \approx -0.2$ (in HR units per second, given our 1 Hz discretization). Thus, the tuned parameters suggest a weak lagged coupling signal from respiration to HR in this segment, even though the overall prediction and filtering RMSE for Model B remained worse than Model A.

8.2 Does this match the project hypothesis?

The project hypothesis was: *if respiration explains additional structure, Model B should reduce innovation magnitude/variance and improve prediction.* In this 30-minute window, the results provide only **partial** support. The grid search selected a *nonzero* respiration coupling term and lag ($b = -0.2$, $\ell = 10$), suggesting a weak lagged association between the PTAF-derived driver and HR dynamics. However, **Model B did not improve predictive performance**: both RMSE metrics were higher than Model A. Therefore, while the tuning found a small lagged respiration contribution, it was not strong or consistent enough (given this driver construction and model form) to reduce overall prediction error or clearly improve innovation behavior compared to the simpler AR(1) baseline.

8.3 Why might respiration not help here?

These outcomes can happen even if true physiology is coupled. In fact, the grid search selected a *nonzero* lagged term ($b = -0.2$ at $\ell = 10$), but this contribution was not strong enough to improve prediction error overall:

- **Driver mismatch:** A simple standardized PTAF-derived driver may not capture the specific respiration features that modulate HR (e.g., respiratory phase, breath timing, or band-limited power related to RSA).
- **Lag/structure mismatch:** Although a best lag of $\ell = 10$ seconds was selected, the true interaction may involve multiple lags, phase-dependent effects, or frequency-specific coupling rather than a single delayed input term.
- **Model simplicity:** A first-order AR/ARX model may be too restrictive to represent oscillatory autonomic dynamics; richer state vectors could model slow baseline plus rhythmic components and allow coupling to vary over time.
- **Segment dependence:** Coupling strength changes with sleep stage and arousals; the chosen 30-minute window may contain heterogeneous physiology (or weak coupling), limiting the benefit of a respiration input.

8.4 Next steps

For next steps we could:

- Upgrade the dynamics to AR(p) or a multi-state oscillator state-space model to capture rhythmic components and slow trends simultaneously.
- Engineer respiration features more aligned with physiology (such as band-limited respiration, phase/amplitude features, or an RSA-oriented proxy) and allow *multi-lag* or frequency-specific coupling.
- Evaluate performance and innovation statistics *stratified by sleep stage/arousal* and across multiple subjects to test whether coupling is stronger in specific conditions or subpopulations.

9 Conclusion

I implemented two Kalman filter formulations for HR dynamics using HSP PSG data: a self predictive AR(1) model and an ARX(1) extension with a PTAF derived respiration driver. In the evaluated 30 minute window, the AR(1) baseline achieved lower prediction and filtering error. While the ARX model selected a *lagged* respiration term ($b = -0.2$ with $\ell = 10$), this did not translate into improved RMSE, suggesting that the added respiration driver (as constructed here) provides at most a weak contribution relative to HR's own persistence. Future work should explore more aligned respiration features and more expressive state space models, ideally evaluated across sleep stages and multiple recordings.